

## 14.7 Near replicates and nonparametric lack-of-fit tests

Exact replicates are a luxury of designed experiments. With continuous regressors, rows of the model matrix are rarely identical: stack loss with all three operating variables has 21 rows and a single repeated pair, one degree of freedom for pure error. The pure-error test survives in a more general form: any larger model chosen without looking at the response gives an exact test, and so do regressors built from the fitted values. What is lost is the guarantee that the larger model contains every alternative: each test looks in some directions and is blind in others.

### 14.7.1 A general recipe

#### Proposition 14.7.1. Testing against any larger model

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\theta}, \sigma^2 \mathbf{I})$ , let  $\mathbf{X}$  have rank  $r$ , and let  $\mathbf{Z}$  be a fixed matrix, chosen without reference to  $\mathbf{Y}$ , with  $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{Z})$  and  $\text{rank}(\mathbf{Z}) = s$ ,  $r < s < n$ . Let  $\mathbf{M}$  and  $\mathbf{M}_Z$  be the two projections and

$$F = \frac{[\text{SSE}(\mathbf{X}) - \text{SSE}(\mathbf{Z})]/(s - r)}{\text{SSE}(\mathbf{Z})/(n - s)}.$$

- (a) If  $\boldsymbol{\theta} \in \mathcal{C}(\mathbf{X})$ , then  $F \sim F(s - r, n - s)$ .
- (b) If  $\boldsymbol{\theta} \in \mathcal{C}(\mathbf{Z})$ , then  $F \sim F(s - r, n - s, \|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2/\sigma^2)$ .
- (c) In general  $F = \frac{U/(s - r)}{V/(n - s)}$  with independent  $U \sim \chi^2(s - r, \gamma_1)$  and  $V \sim \chi^2(n - s, \gamma_2)$ , where  $\gamma_1 = \|(\mathbf{M}_Z - \mathbf{M})\boldsymbol{\theta}\|^2/\sigma^2$  and  $\gamma_2 = \|(\mathbf{I} - \mathbf{M}_Z)\boldsymbol{\theta}\|^2/\sigma^2$ . For every  $c > 0$ ,  $\Pr(F > c)$  is strictly increasing in  $\gamma_1$  and strictly decreasing in  $\gamma_2$ .

*Proof.* The decomposition  $\mathbf{I} - \mathbf{M} = (\mathbf{M}_Z - \mathbf{M}) + (\mathbf{I} - \mathbf{M}_Z)$  into orthogonal projections of ranks  $s - r$  and  $n - s$  comes from Theorem 6.5.1, and Theorem 4.4.4 gives (c) apart from the monotonicity. Parts (a) and (b) are the cases  $\gamma_1 = \gamma_2 = 0$  and  $\gamma_2 = 0$ . For the monotonicity, write  $\Pr(F > c) = E[\Pr(U > c'V \mid V)]$  with  $c' = c(s - r)/(n - s)$ . For each value  $V > 0$  the inner probability is strictly increasing in  $\gamma_1$  by Proposition 4.1.3. Similarly  $\Pr(F > c) = E[\Pr(V < U/c' \mid U)]$ , and  $\Pr\{\chi^2(n - s, \gamma_2) < u\}$  is strictly decreasing in  $\gamma_2$  for  $u > 0$ . ■

The size is exact however  $\mathbf{Z}$  is chosen without using  $\mathbf{Y}$ ; with random regressors,  $\mathbf{Z}$  may be any function of  $\mathbf{X}$  (Theorem 14.1.2). The pure-error test is the case  $\mathbf{Z} = \text{cell means}$ . Part (c) describes what happens under the alternatives that matter, where the true mean lies in neither model. Departures captured by the larger model raise  $\gamma_1$  and help; departures outside it raise  $\gamma_2$ , inflate the denominator and hurt. A departure orthogonal to  $\mathcal{C}(\mathbf{Z})$  gives power *below* the level (Exercise (B3)).

## 14.7.2 Near replicates, partitions and the rainbow test

Without exact replicates, the natural substitute is to group rows that are *nearly* identical. A cluster analysis of the standardized rows of  $\mathbf{X}$  forms  $g$  clusters of near replicates, with indicator matrix  $\mathbf{D}$ . Testing  $\mathbf{X}$  against  $\mathbf{Z} = [\mathbf{X}, \mathbf{D}]$  allows each cluster its own shift from the fitted plane. This is the near-replicate test studied by Christensen (1989); Shillington (1979) proposed a variant based on the cluster averages of the regressors, and Christensen (1991) characterized the kinds of lack of fit each detects. With exact replicates all reduce to the pure-error test.

A second family partitions the cases instead of the regressor space. If the rows are split into two groups, with model matrices  $\mathbf{X}_1$  and  $\mathbf{X}_2$ , the model that fits a separate regression to each group has  $\mathbf{Z} = \begin{pmatrix} \mathbf{X}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{X}_2 \end{pmatrix}$ , whose column space contains  $\mathcal{C}(\mathbf{X})$ , and  $\text{SSE}(\mathbf{Z})$  is the sum of the two residual sums of squares (Exercise (B1)). A straight line fitted to a curved mean is beaten by two lines, one for each half of the range. Utts (1982) proposed an extreme partition, the **rainbow test**: fit the model to a central subset of  $m$  cases, those with the smallest leverages, and give each of the remaining  $n - m$  cases a free mean of its own. Then  $\text{SSE}(\mathbf{Z})$  is the residual sum of squares of the central fit, and the test asks whether the model fitted to the middle of the design extrapolates to its edges.

All of these are exact by Proposition 14.7.1, and all require a decision made from  $\mathbf{X}$  alone: how many clusters, where to split, how large a central subset. Choosing the clusters or the split after looking at the residuals destroys the exactness, because  $\mathbf{Z}$  then depends on  $\mathbf{Y}$ .

## 14.7.3 Richer mean functions

The most direct alternative is a larger parametric model: add squares and cross-products of the regressors, or a regression spline. For a single regressor, a cubic spline with interior knots  $\kappa_1, \dots, \kappa_m$  adds the columns  $x^2$ ,  $x^3$  and  $(x - \kappa_l)_+^3$ ,  $l = 1, \dots, m$ , and can follow any smooth curve if the knots are dense enough. These are reduced-model  $F$  tests (Theorem 11.1.1), exact by Proposition 14.7.1(a) provided the larger model is fixed before seeing the data. Their power is concentrated on departures that the added columns can represent. A quadratic term is excellent against smooth curvature and nearly useless against a departure confined to a small part of the range.

## 14.7.4 Regressors built from the fitted values

A classical alternative looks for departures along the fitted values: if the mean is a nonlinear function of  $\mathbf{x}^\top \boldsymbol{\beta}$ , then powers of  $\hat{y}$  should help predict  $y$ . But  $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{Y}$  depends on the response, so Proposition 14.7.1 does not apply. Remarkably, the  $F$  test is exact anyway.

### Theorem 14.7.2. Regressors that are functions of the fitted values

Let  $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$  with  $\text{rank}(\mathbf{X}) = r$ . Let  $\mathbf{W} = w(\mathbf{M}\mathbf{Y})$  be an  $n \times q$  matrix computed from the fitted values alone, with  $\text{rank}[\mathbf{X}, \mathbf{W}] = r + q < n$  with probability one. Then

$$F = \frac{[\text{SSE}(\mathbf{X}) - \text{SSE}([\mathbf{X}, \mathbf{W}])]/q}{\text{SSE}([\mathbf{X}, \mathbf{W}])/(n - r - q)} \sim F(q, n - r - q).$$

*Proof.* Let  $\tilde{\mathbf{W}} = (\mathbf{I} - \mathbf{M})\mathbf{W}$ , which has rank  $q$ , and let  $\mathbf{P}$  project onto  $\mathcal{C}(\tilde{\mathbf{W}})$ . As in Lemma 6.6.1,  $\mathcal{C}([\mathbf{X}, \mathbf{W}]) = \mathcal{C}(\mathbf{X}) \oplus \mathcal{C}(\tilde{\mathbf{W}})$  with orthogonal summands, so the projection onto  $\mathcal{C}([\mathbf{X}, \mathbf{W}])$  is  $\mathbf{M} + \mathbf{P}$ . With  $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\mathbf{Y}$ , and since  $\mathbf{P}\mathbf{M} = \mathbf{0}$ ,

$$\begin{aligned} \text{SSE}(\mathbf{X}) - \text{SSE}([\mathbf{X}, \mathbf{W}]) &= \|\mathbf{P}\mathbf{Y}\|^2 = \|\mathbf{P}\hat{\boldsymbol{\varepsilon}}\|^2, \\ \text{SSE}([\mathbf{X}, \mathbf{W}]) &= \|\hat{\boldsymbol{\varepsilon}}\|^2 - \|\mathbf{P}\hat{\boldsymbol{\varepsilon}}\|^2. \end{aligned}$$

By Theorem 4.4.4,  $\mathbf{M}\mathbf{Y}$  and  $\hat{\boldsymbol{\varepsilon}}$  are independent, and  $\hat{\boldsymbol{\varepsilon}} \sim N_n(\mathbf{0}, \sigma^2(\mathbf{I} - \mathbf{M}))$ . So, given  $\mathbf{M}\mathbf{Y}$ , the matrix  $\mathbf{P}$  is fixed and  $\hat{\boldsymbol{\varepsilon}}$  keeps its distribution, which is that of  $(\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}^*$  for  $\boldsymbol{\varepsilon}^* \sim N_n(\mathbf{0}, \sigma^2\mathbf{I})$ . Because  $\mathcal{C}(\mathbf{P}) \subseteq \mathcal{C}(\mathbf{X})^\perp$ ,  $\mathbf{P}(\mathbf{I} - \mathbf{M}) = \mathbf{P}$ , and the two sums of squares become  $\|\mathbf{P}\boldsymbol{\varepsilon}^*\|^2$  and  $\|(\mathbf{I} - \mathbf{M} - \mathbf{P})\boldsymbol{\varepsilon}^*\|^2$ . These are quadratic forms in orthogonal projections of ranks  $q$  and  $n - r - q$ , independent  $\sigma^2\chi^2(q)$  and  $\sigma^2\chi^2(n - r - q)$  variables by Theorem 4.4.4. The conditional law of  $F$  is therefore  $F(q, n - r - q)$  whatever  $\mathbf{M}\mathbf{Y}$  is, and by Lemma 14.1.1 it is also the unconditional law. ■

The theorem is due to Milliken and Graybill (1970). Its best-known case is Ramsey's (1969) **RESET**, which adds the squares and cubes of the fitted values, entry by entry. Tukey's (1949) one degree of freedom for nonadditivity, which tests a two-way additive layout against a multiplicative interaction, can be written in the same form with the squared fitted values. With a single regressor the construction brings nothing new: when  $\hat{\beta}_1 \neq 0$ , the columns  $\hat{\mathbf{y}}^2$  and  $\hat{\mathbf{y}}^3$  span, together with  $\mathbf{1}$  and  $\mathbf{x}$ , the same space as  $x^2$  and  $x^3$ , so RESET is the cubic polynomial test (Exercise (B4)). Its value is in many dimensions, where it spends two degrees of freedom on the one direction along which the fit varies.

### 14.7.5 Smoothing-based tests

Instead of committing to a parametric alternative, one can ask whether the residuals contain any smooth structure. Let  $\mathbf{S}$  be a fixed smoother matrix, for instance kernel weights that average each residual with those of its neighbours in the regressor space, and consider

$$T = \frac{\|\mathbf{S}\hat{\boldsymbol{\varepsilon}}\|^2}{\|\hat{\boldsymbol{\varepsilon}}\|^2}. \quad (14.7.1)$$

If the model is right, the residuals are noise and smoothing them leaves little; if the mean has structure the model misses, the smoothed residuals keep it and  $T$  is large.

#### Proposition 14.7.3. The smoother statistic is pivotal

If  $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$ , the distribution of  $T$  in Eq. 14.7.1 depends only on  $\mathbf{X}$  and  $\mathbf{S}$ , not on  $\boldsymbol{\beta}$  or  $\sigma^2$ . It equals the distribution of  $\|\mathbf{S}(\mathbf{I} - \mathbf{M})\mathbf{u}\|^2 / \|(\mathbf{I} - \mathbf{M})\mathbf{u}\|^2$  with  $\mathbf{u} \sim N_n(\mathbf{0}, \mathbf{I})$ .

*Proof.*  $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\mathbf{Y} = (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}$ , and  $\boldsymbol{\varepsilon} = \sigma\mathbf{u}$  with  $\mathbf{u} \sim N_n(\mathbf{0}, \mathbf{I})$ . The factor  $\sigma^2$  cancels in the ratio. ■

So the null distribution of  $T$  can be simulated as precisely as desired, from the design alone, and the test has exact size apart from Monte Carlo error. Statistics of this kind were proposed by Azzalini and Bowman (1993), who compare the residual sums of squares of the parametric fit and of a nonparametric smoother, and they are developed at length by Hart (1997). The power depends on the amount of smoothing, and a bandwidth chosen after inspecting the data breaks the exact size. With several regressors, neighbourhoods become sparse, so in practice one smooths along a single direction, such as the fitted values. And exactness depends, as for every test in this section, on normal homoscedastic errors.

## 14.7.6 Stack loss with all three regressors

### Example. Is the three-variable model adequate?

Return to the stack loss data with air flow, water temperature and acid concentration,  $n = 21$  and  $r = 4$ . Ward clustering of the standardized rows into 8 clusters and testing  $\mathbf{X}$  against  $[\mathbf{X}, \mathbf{D}]$  gives  $F = 1.43$  on 7 and 10 degrees of freedom ( $p$ -value 0.292). Adding the squares of the three regressors gives  $F = 1.92$  on 3 and 14 ( $p$ -value 0.172). The rainbow test with the 11 cases of smallest leverage as the central subset gives  $F = 1.04$  on 10 and 7 ( $p$ -value 0.497), and RESET gives  $F = 1.17$  on 2 and 15 ( $p$ -value 0.336). None of the tests finds lack of fit. The strong lack of fit of the line in air flow alone (Example (Stack loss and air flow)) was the omitted water temperature, which varies within the groups of equal air flow. With 21 cases none of these tests is powerful, and a few unusual days deserve the scrutiny of [Chapter 20](#).

```
import numpy as np
import statsmodels.api as sm
from scipy import stats
from scipy.cluster.hierarchy import fcluster, linkage
from statsmodels.stats.diagnostic import linear_reset

def proj(A):
    """Orthogonal projection onto  $C(A)$ , via an orthonormal basis of the column space
    ."""
    U, s, _ = np.linalg.svd(A, full_matrices=False)
    U = U[:, s > s[0] * 1e-10]
    return U @ U.T

def f_test(y, X, Zbig):
    """F test of  $C(X)$  against a larger  $C(Zbig) \supseteq C(X)$ ; returns  $F$ ,  $df1$ ,  $df2$ ,  $p$ ."""
    n = len(y)
    M, MZ = proj(X), proj(Zbig)
    r, s = round(np.trace(M)), round(np.trace(MZ))
    sse, sse_z = y @ (y - M @ y), y @ (y - MZ @ y)
    F = ((sse - sse_z) / (s - r)) / (sse_z / (n - s))
    return F, s - r, n - s, stats.f.sf(F, s - r, n - s)

data = sm.datasets.stackloss.load_pandas().data
y = data["STACKLOSS"].to_numpy()
R = data[["AIRFLOW", "WATERTEMP", "ACIDCONC"]].to_numpy()
n = len(y)
X = np.column_stack([np.ones(n), R])
fit = sm.OLS(y, X).fit()

# exact replicates: rows of X that occur more than once
_, rows = np.unique(R, axis=0, return_inverse=True)
```

```

print("distinct rows:", rows.max() + 1, "of", n)

# near replicates: cluster the standardized regressor rows (Ward linkage), c
clusters
Rs = (R - R.mean(axis=0)) / R.std(axis=0)
clusters = fcluster(linkage(Rs, method="ward"), t=8, criterion="maxclust")
D = np.eye(clusters.max())[clusters - 1] # cluster indicators
tests = {"near replicates": f_test(y, X, np.column_stack([X, D])),
        "added squares": f_test(y, X, np.column_stack([X, R ** 2]))}

# rainbow: refit on the half of the cases with the smallest leverages
h = np.diag(proj(X))
central = np.argsort(h)[: n // 2 + 1]
E = np.eye(n)[:, np.setdiff1d(np.arange(n), central)] # one free mean per outer
case
tests["rainbow"] = f_test(y, X, np.column_stack([X, E]))

# RESET: add squares and cubes of the fitted values
reset = linear_reset(fit, power=3, test_type="fitted", use_f=True)
tests["RESET"] = (reset.fvalue, reset.df_num, reset.df_denom, reset.pvalue)
for name, (F, d1, d2, p) in tests.items():
    print(f"{name:16s} F = {F:6.2f} on ({d1:.0f}, {d2:.0f}) df, p = {p:.3f}")

```

[Run this code online](#)

## 14.7.7 Which test?

No lack-of-fit test is best against every alternative: by [Proposition 14.7.1\(c\)](#), a test gains power from departures its larger model can represent and loses power from the others.

### Example. Two departures, five tests

Take  $n = 50$  equally spaced values of  $x$  in  $(0, 1)$ , a straight-line working model, standard normal errors, and five tests at level 0.05: adding  $x^2$ ; RESET with squared and cubed fitted values; a cubic spline with knots at 0.25, 0.5 and 0.75; near replicates formed by 10 clusters of 5 neighbouring values; and the smoother statistic [Eq. 14.7.1](#) with Gaussian kernel weights of bandwidth 0.06, its critical value simulated from 50000 null residual vectors. Over 4000 samples from the straight line, the rejection rates are 0.050, 0.054, 0.054, 0.052 and 0.054: all five tests are exact.

Against smooth curvature, a mean of  $1 + 2x + 6(x - 0.5)^2$ , the powers are 0.874 for the quadratic term, 0.786 for RESET (which here is the cubic polynomial test), 0.606 for the spline, 0.422 for near replicates and 0.723 for the smoother. Against a narrow bump of height 2.5 near  $x = 0.8$ , the order is reversed: the smoother has power 0.622, the spline 0.501, near replicates 0.459, RESET 0.227, and the quadratic term 0.029, *below* the level: the bump is nearly orthogonal to  $x^2$  after the line is removed, so it inflates the denominator of that test without feeding its numerator. [Figure 14.7.1](#) shows the power curves.

```

rng = np.random.default_rng(1471)
n2, reps = 50, 4000
x = (np.arange(n2) + 0.5) / n2 # fixed design on (0, 1)
X2 = np.column_stack([np.ones(n2), x])
M2 = proj(X2)
knots = (0.25, 0.5, 0.75)

```

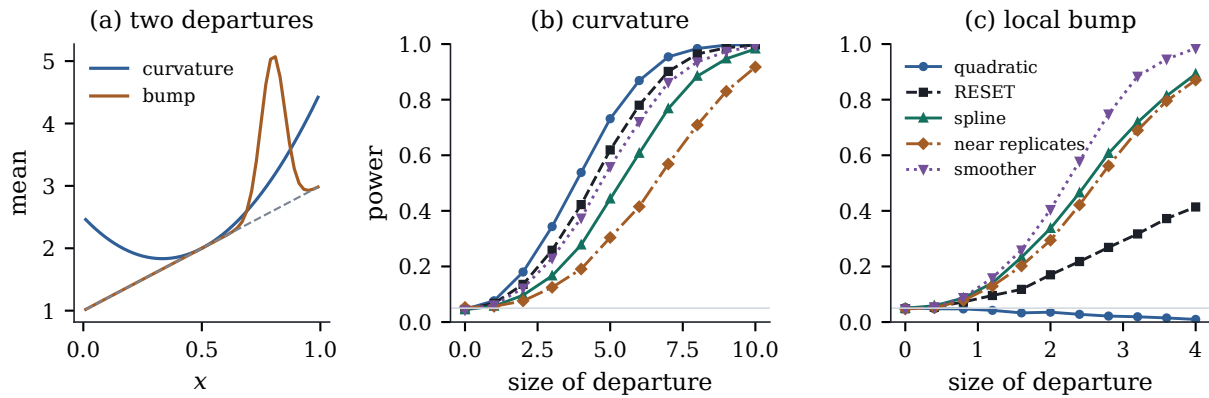


Figure 14.7.1: Power of five exact lack-of-fit tests of a straight line ( $n = 50$ , level 0.05). (a) The two departures. (b) Smooth curvature. (c) A local bump, against which the quadratic test has power below its level.

```
spline = np.column_stack([x ** 2, x ** 3] + [np.clip(x - k, 0, None) ** 3 for k in
    knots])
cells = np.eye(10)[np.arange(n2) // 5] # 10 clusters of 5
    neighbouring x
bigger = {"quadratic": np.column_stack([X2, x ** 2]),
    "spline": np.column_stack([X2, spline]),
    "near replicates": np.column_stack([X2, cells])}
proj_big = {k: proj(v) for k, v in bigger.items()}

# kernel smoother of the residuals;  $T = |S e|^2 / |e|^2$ , null law free of beta and
    sigma
bw = 0.06
K = np.exp(-0.5 * ((x[:, None] - x[None, :]) / bw) ** 2)
S = K / K.sum(axis=1, keepdims=True)
null_e = (np.eye(n2) - M2) @ rng.normal(size=(n2, 50_000))
T_null = np.sum((S @ null_e) ** 2, axis=0) / np.sum(null_e ** 2, axis=0)
T_crit = np.quantile(T_null, 0.95)

def powers(mean):
    """Rejection rates at level 0.05 of each test when  $E(Y) = \text{mean}$ ."""
    Y = mean[:, None] + rng.normal(size=(n2, reps))
    e = Y - M2 @ Y
    sse = np.sum(e ** 2, axis=0)
    out = {}
    for name, P in proj_big.items():
        s = round(np.trace(P))
        sse_z = np.sum((Y - P @ Y) ** 2, axis=0)
        F = ((sse - sse_z) / (s - 2)) / (sse_z / (n2 - s))
        out[name] = np.mean(F > stats.f.ppf(0.95, s - 2, n2 - s))
    Fr = np.empty(reps)
    for j in range(reps): # RESET: regressors depend on
        y
        yh = M2 @ Y[:, j]
        W = np.column_stack([yh ** 2, yh ** 3])
        Wt = W - M2 @ W
        g, *_ = np.linalg.lstsq(Wt, e[:, j], rcond=None)
        ss_add = e[:, j] @ (Wt @ g)
        Fr[j] = (ss_add / 2) / ((sse[j] - ss_add) / (n2 - 4))
    out["RESET"] = np.mean(Fr > stats.f.ppf(0.95, 2, n2 - 4))
    T = np.sum((S @ e) ** 2, axis=0) / sse
```

```

    out["smoother"] = np.mean(T > T_crit)
    return out

curve = (x - 0.5) ** 2                                # smooth curvature
bump = np.exp(-0.5 * ((x - 0.8) / 0.05) ** 2)         # a local bump
sizes = powers(1 + 2 * x)
print("sizes:", {k: round(float(v), 3) for k, v in sizes.items()})
reported = {}
for label, shape, a in [("curvature", curve, 6.0), ("bump", bump, 2.5)]:
    reported[label] = powers(1 + 2 * x + a * shape)
    print(f"{label:9s}:", {k: round(float(v), 3) for k, v in reported[label].items()})

```

[Run this code online](#)

The lesson is to decide which inadequacy matters before testing: a low-degree polynomial or RESET for smooth curvature, a flexible alternative or a smoother for local departures. Only the pure-error test of Theorem 14.6.2 has a larger model containing every mean function of the regressors. Residual plots, which look in all directions informally, remain indispensable ([Chapter 20](#)).

## 14.7.8 Exercises

### 14.7.8.1 A. Check your understanding

#### Exercise. A1

A regression with an intercept and two regressors has  $n = 30$  cases, whose rows are grouped into 8 clusters of near replicates. Assuming no combination of cluster indicators lies in  $\mathcal{C}(\mathbf{X})$  except the constant, give the degrees of freedom of the near-replicate test of  $\mathbf{X}$  against  $[\mathbf{X}, \mathbf{D}]$ .

### 14.7.8.2 B. Practice

#### Exercise. B1

Let  $\mathbf{X} = \begin{pmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{pmatrix}$  and  $\mathbf{Z} = \begin{pmatrix} \mathbf{X}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{X}_2 \end{pmatrix}$ . Show that  $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{Z})$ , that the projection onto  $\mathcal{C}(\mathbf{Z})$  is  $\begin{pmatrix} \mathbf{M}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_2 \end{pmatrix}$ , and that  $\text{SSE}(\mathbf{Z})$  is the sum of the residual sums of squares of the two separate regressions.

*Solution.*  $\mathbf{X} = \mathbf{Z} \begin{pmatrix} \mathbf{I} \\ \mathbf{0} \end{pmatrix}$ , so  $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{Z})$ . The block-diagonal matrix of the two projections is symmetric and idempotent, and its column space is  $\mathcal{C}(\mathbf{X}_1) \times \mathcal{C}(\mathbf{X}_2) = \mathcal{C}(\mathbf{Z})$ , so it is the projection onto  $\mathcal{C}(\mathbf{Z})$  (Theorem 6.3.3). Applying  $\mathbf{I}$  minus it to  $\mathbf{y} = (\mathbf{y}_1^\top, \mathbf{y}_2^\top)^\top$  gives the two residual vectors  $(\mathbf{I} - \mathbf{M}_1)\mathbf{y}_1$  and  $(\mathbf{I} - \mathbf{M}_2)\mathbf{y}_2$ , whose squared lengths add.

#### Exercise. B2

In the rainbow test, order the cases so that the central subset of size  $m$  comes first, and let its model matrix  $\mathbf{X}_c$  have rank  $r$ . Show that  $\mathbf{Z} = \begin{pmatrix} \mathbf{X}_c & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_{n-m} \end{pmatrix}$  has  $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{Z})$ , that  $\text{SSE}(\mathbf{Z})$  is the residual sum of squares of the central fit, and that the test has  $n - m$  and  $m - r$  degrees

of freedom.

### Exercise. B3

In [Proposition 14.7.1](#), suppose  $\boldsymbol{\theta} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\delta}$  with  $\boldsymbol{\delta} \neq \mathbf{0}$  orthogonal to  $\mathcal{C}(\mathbf{Z})$ . Show that the power of the test is strictly less than its size.

*Solution.* Here  $(\mathbf{M}_Z - \mathbf{M})\boldsymbol{\theta} = \mathbf{M}_Z\boldsymbol{\delta} - \mathbf{M}\boldsymbol{\delta} = \mathbf{0}$ , since  $\boldsymbol{\delta} \perp \mathcal{C}(\mathbf{Z}) \supseteq \mathcal{C}(\mathbf{X})$ , so  $\gamma_1 = 0$ , while  $\gamma_2 = \|\boldsymbol{\delta}\|^2/\sigma^2 > 0$ . By part (c),  $\Pr(F > c)$  is strictly smaller than its value at  $\gamma_2 = 0$ , which is the size by part (a).

### Exercise. B4

With a single regressor and an intercept, show that if  $\hat{\beta}_1 \neq 0$ , then  $\mathcal{C}([\mathbf{1}, \mathbf{x}, \hat{\mathbf{y}}^2, \hat{\mathbf{y}}^3]) = \mathcal{C}([\mathbf{1}, \mathbf{x}, \mathbf{x}^2, \mathbf{x}^3])$ , powers taken entry by entry. Conclude that RESET is then the  $F$  test of the cubic polynomial.

#### 14.7.8.3 C. Going deeper

### Exercise. C1

In a two-way layout with one observation per cell and the additive model  $E(y_{ij}) = \mu + \alpha_i + \beta_j$ , Tukey's statistic tests the added regressor  $(\hat{\alpha}_i \hat{\beta}_j)$ . Show that, after the additive model is projected out, this regressor spans the same space as the entrywise square of the fitted values, and deduce from [Theorem 14.7.2](#) that Tukey's test has an exact  $F(1, (a-1)(b-1)-1)$  null distribution.